

Methodology for Measuring Intrinsic EMC-Relevant Emissions of Power Semiconductor Devices

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Vorgelegt von

Robert Kragl

aus Frankfurt am Main

Hauptberichter: Univ.-Prof. Dr. Ingmar Kallfass
Mitberichter: XXXXX
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Institut für Robuste Leistungshalbleitersysteme der Universität Stuttgart

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Abstract

This thesis examines the interface between two domains of power electronics that are frequently treated as distinct, despite their fundamental interdependence. Although semiconductor device switching is the primary source of conducted and radiated emissions, electromagnetic compatibility (EMC) continues to be addressed predominantly as a system-level problem rather than as an intrinsic semiconductor design variable. This disconnect stems from a fundamental measurement limitation: conventional characterization methods of power semiconductor devices are unable to resolve the high-frequency, EMC-relevant signal components emitted by the device, as these components fall below the measurement noise floor. As a quantity cannot be optimized without first being measurable, this absence of characterization has prevented electromagnetic noise from being treated as an intrinsic semiconductor design target. Consequently, noise optimization is often deferred to later design stages, where mitigation is instead implemented through converter- or system-level measures, at the expense of efficiency, compactness, and development time.

The contribution of this thesis is a measurement methodology that enables direct, low-noise characterization of intrinsic electromagnetic emissions at the semiconductor power terminals. This is achieved through the development of a dual-channel approach for both a differential active voltage and a current probe, combined with post-processing methods for residual-noise suppression. Together, these enable semiconductor-focused EMC analysis up to 400 MHz while preserving the measurement of conventional performance indicators such as switching losses, overvoltage, and dv/dt . An additional event-resolved spectral analysis method provides deeper insight into individual noise contributions. Because this methodology renders a previously unobservable behavior measurable for the first time, particular emphasis is placed on verifying the measurement setup.

Applying this methodology reveals broadband differences between SiC MOSFET variants and operating points, quantifying effects that were previously only assumed to be relevant. It further links switching behavior directly to EMC-relevant spectral components, establishing device emissions as a measurable design target. The developed approach is not limited to a specific technology and provides a general basis for semiconductor-focused EMC analysis at multiple stages of the development of power electronic systems: (1) during device technology development, it enables the characterization of different design variants, allowing direct evaluation of trade-offs with conventional performance metrics; (2) during converter design, it enables the benchmarking of candidate semiconductor devices with respect to their electromagnetic emissions for a given application; and (3) during compliance testing, it enables the identification of the switching event most likely to dominate emissions and cause violations of the applicable standard limits. Collectively, these contributions support earlier, EMC-aware design decisions across future power semiconductor and converter development.

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